Spectral analysis II

Version 2 (April 8, 2026)

ECON 31730, Spring 2026

University of Chicago
Department of Economics

Mikkel Plagborg-Moller

Autoregressive spectral estimation

- We continue with univariate series for now.
- We know any cts spectrum can be approximated by an AR(p) model.
- Most straight-forward estimator of the spectrum:

$$\hat{s}_y^{(p)}(\omega) = \frac{\hat{\sigma}^2}{2\pi} \left| 1 - \sum_{\ell=1}^{p} \hat{A}_\ell e^{-i\omega\ell} \right|^{-2}, \quad \omega \in [0, \pi],$$

  where $(\hat{A}_1, \dots, \hat{A}_p)$ and $\hat{\sigma}^2$ are the least-squares AR(p) estimators.
- If the true model were an AR(p), $\hat{s}_y^{(p)}(\omega)$ would clearly be consistent at every ω.
- Could select p by AIC/BIC. Could also report results for different values of p.

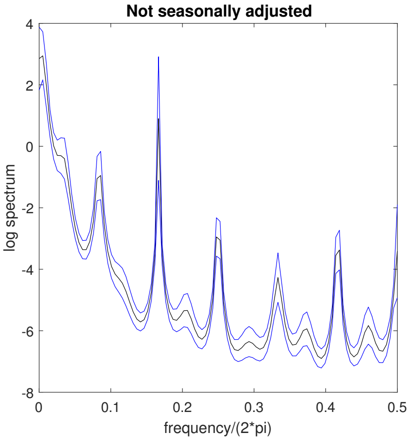
Autoregressive spectral estimation: confidence band

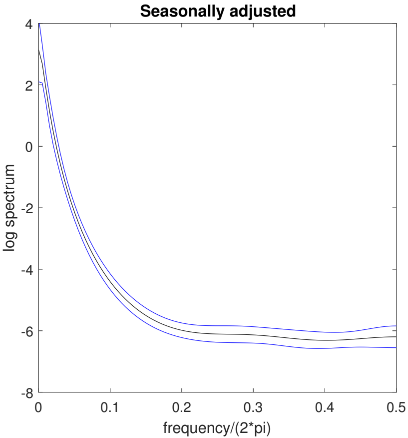
- Let's suppose the true model is an AR(p). How to compute standard errors for $\hat{s}_y^{(p)}(\omega)$ at each frequency ω?
- Easy: delta method.
- Customary and convenient to report s.e. for the *log* spectrum

$$\log \hat{s}_y^{(p)}(\omega) = \log \hat{\sigma}^2 - \log(2\pi) - \log \left| 1 - \sum_{\ell=1}^{p} \hat{A}_\ell e^{-i\omega\ell} \right|^2, \;\; \omega \in [0, \pi].$$

- Take derivatives wrt. σ^2 and A_ℓ ($1 \leq \ell \leq p$) to apply delta method.
- A plot of confidence intervals for each value $s_y(\omega)$, over a grid of frequencies $\omega \in [0, \pi]$, is called a **(pointwise) confidence band**.
- "Pointwise": Confidence statement holds at each ω separately. But band does not cover *entire* true spectral density with prob. $1 - \alpha$.

Example: AR conf. band for unempl. rate spectrum





Sample: 1948:1–2017:8 (U.S.). Pointwise confidence level: 95%.
AR lag length: 25 (left), 5 (right).

Autoregressive spectral estimation: consistency

- What if the true model is an AR(∞)? (Almost WLOG.)

Proposition (Berk, 1974, Thm. 1)

Let $\{y_t\}$ be a cov. stat. univariate AR(∞) process $A(L)y_t = u_t$, where $u_t \overset{iid}{\sim} (0, \sigma^2)$, $E(u_t^4) < \infty$. Assume $A(z) = 1 - \sum_{\ell=1}^{\infty} A_\ell z^\ell$ has all roots outside the unit circle.

Fix $\omega \in [0, \pi]$, and let $\hat{s}_y^{(p)}(\omega)$ denote the least-squares AR(p) estimator of the true $s_y(\omega)$. Choose p_T such that:

i) $p_T^3/T \to 0$ as $T \to \infty$.

ii) $\sqrt{p_T} \sum_{\ell=p_T+1}^{\infty} |A_\ell| \to 0$ as $T \to \infty$.

Then $\hat{s}_y^{(p_T)}(\omega) \overset{p}{\to} s_y(\omega)$ as $T \to \infty$.

- Get *nonparametric* consistency of AR(p_T) estimator, provided $p_T \to \infty$ at rate slower than $T^{1/3}$ and $A_\ell \to 0$ quickly as $\ell \to \infty$.

Autoregressive spectral estimation: Bayesian approach

- It is straight-forward to do Bayesian inference on the spectrum of a Gaussian AR(p) model.

  1. Draw from the posterior distribution of the AR parameters.
  2. Transform draw into spectral density $s_y(\omega)$ (perhaps for several different ω values), using AR formula.
  3. Repeat many times.
  4. Plot posterior means/medians/quantiles.

- Can answer questions like: "What is the probability (conditional on the data) that the spectral density is monotonically decreasing?"

Periodogram

- Although AR spectral estimation is nonparametrically consistent, in any finite sample results could be sensitive to the chosen lag order.
- There is another explicitly nonparametric approach to estimation.
- Set of **Fourier frequencies**: $\mathcal{F}_T \equiv \{\omega \in (-\pi, \pi] \colon \omega = \frac{2\pi j}{T},\ j \in \mathbb{Z}\}$.

Definition

The **periodogram** at **ordinate** j is given by

$$\hat{\mathcal{I}}(\omega_j) \equiv \left| T^{-1/2} \sum_{t=1}^{T} e^{-i\omega_j t} y_t \right|^2, \quad \omega_j \equiv \frac{2\pi j}{T} \in \mathcal{F}_T.$$

The quantity inside the abs. value is the **discrete Fourier transform** of the data at ordinate j.

- Exercise: Show that, at any non-zero frequency, the periodogram is unaffected by shifting all data by a constant: $y_t + c$, $t = 1, \ldots, T$.

Periodogram (cont.)

- The periodogram is a sample analogue of $(2\pi\times)$ the spectral density.
- Define $\mu_y \equiv E[y_t]$. At any frequency $\omega_j \neq 0$,

$$
\begin{aligned}
\hat{\mathcal{I}}(\omega_j) &= \frac{1}{T}\sum_{t=1}^{T} e^{-i\omega_j t} y_t \sum_{s=1}^{T} e^{i\omega_j s} y_s \\
&= \frac{1}{T}\sum_{t=1}^{T} e^{-i\omega_j t}(y_t - \mu_y) \sum_{s=1}^{T} e^{i\omega_j s}(y_s - \mu_y) \\
&= \frac{1}{T}\sum_{t=1}^{T}\sum_{s=1}^{T} e^{-i\omega_j (t-s)}(y_t - \mu_y)(y_s - \mu_y) \\
&= \sum_{|\ell|<T} e^{-i\omega_j \ell}\tilde{\Gamma}_y(|\ell|),
\end{aligned}
$$

  where $\tilde{\Gamma}_y(\ell) \equiv \frac{1}{T}\sum_{t=1}^{T-\ell}(y_{t+\ell} - \mu_y)(y_t - \mu_y)$.
- At frequency 0, $\hat{\mathcal{I}}(0) = T\hat{\mu}_y^2$, where $\hat{\mu}_y$ is the sample avg.

Periodogram: mean

- Let $g_T(\omega)$ denote the Fourier frequency closest to ω, given T.
- If the true ACF is abs. summable, then for any fixed $\omega \neq 0$,

$$\begin{aligned} E\left[\hat{\mathcal{I}}(g_T(\omega))\right] &= \sum_{|\ell|<T} e^{-ig_T(\omega)\ell} E\left[\tilde{\Gamma}_y(|\ell|)\right] \\ &= \sum_{|\ell|<T} e^{-ig_T(\omega)\ell} \frac{T-|\ell|}{T} \Gamma_y(\ell) \\ &\to \sum_{\ell=-\infty}^{\infty} e^{-i\omega\ell} \Gamma_y(\ell) \quad \text{as } T \to \infty \\ &= 2\pi s_y(\omega), \end{aligned}$$

  using the dominated convergence theorem.
- So $\frac{1}{2\pi}\hat{\mathcal{I}}(g_T(\omega))$ is an asymptotically unbiased estimator of $s_y(\omega)$.
- At frequency 0, there is an asy. bias: $E[\hat{\mathcal{I}}(g_T(0))] - T\mu_y^2 \to 2\pi s_y(0)$.

Periodogram: covariances

- Although the normalized periodogram is unbiased for the spectral density, it is not consistent: The variance does not vanish.
- Under regularity conditions (Brockwell & Davis, 1991, Thm. 10.3.2),

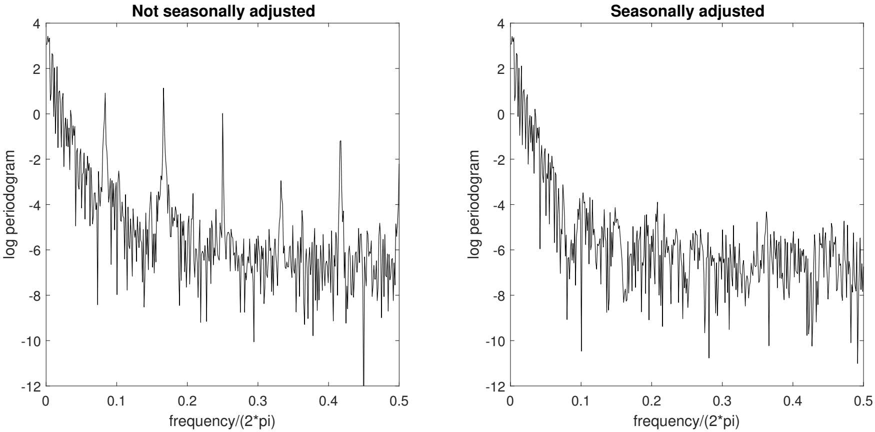
$$\mathrm{Var}(\hat{\mathcal{I}}(\omega_j)) = \{1 + \mathbb{1}(\omega_j \in \{0, \pi\})\}(2\pi)^2 s_y^2(\omega_j) + O(T^{-1/2}).$$

- Intuitively, $\hat{\mathcal{I}}(\omega_j)$ is a sum over a growing number T of sample ACFs. Though each of the sample ACFs is consistent, the sum is not.
- A useful result is that (see above B&D thm.)

$$\mathrm{Cov}(\hat{\mathcal{I}}(\omega_j), \hat{\mathcal{I}}(\omega_k)) = O(T^{-1}), \quad \omega_j \neq \omega_k.$$

- Hence, the periodogram transforms the dependent data $\{y_t\}_{t=1,\dots,T}$ into T approximately uncorrelated (but heteroskedastic) periodogram ordinates $\{\hat{\mathcal{I}}(\omega)\}_{\omega \in \mathcal{F}_T}$.

Example: Periodogram of unemployment rate



Sample: 1948:1–2017:8 (U.S.).

Periodogram: asymptotic distribution

- We can always write

$$\hat{\mathcal{I}}(\omega_j) = \tfrac{1}{2}(\hat{\alpha}(\omega_j)^2 + \hat{\beta}(\omega_j)^2),$$

$$\hat{\alpha}(\omega_j) \equiv \sqrt{\tfrac{2}{T}} \textstyle\sum_{t=1}^T \cos(\omega_j t) y_t, \quad \hat{\beta}(\omega_j) \equiv \sqrt{\tfrac{2}{T}} \textstyle\sum_{t=1}^T \sin(\omega_j t) y_t.$$

- If $y_t \overset{iid}{\sim} N(0,1)$, then $\{\hat{\alpha}(\omega), \hat{\beta}(\omega)\}_{\omega \in \mathcal{F}_T \cap \mathbb{R}_{++}}$ are mutually independent std. normally distributed. This is because one can show

$\sum_{t=1}^T \cos(\omega_j t)\cos(\omega_k t) = \sum_{t=1}^T \sin(\omega_j t)\cos(\omega_k t) = \sum_{t=1}^T \sin(\omega_j t)\sin(\omega_k t) = 0$

when $\omega_j \neq \pm\omega_k$, and $\sum_{t=1}^T \cos(\omega_j t)^2 = \sum_{t=1}^T \sin(\omega_j t)^2 = \frac{T}{2}$ for $\omega_j \neq 0$.

- Hence, when $y_t \overset{iid}{\sim} N(0, \sigma^2)$, we have for all $\omega_j \in \mathcal{F}_T \cap \mathbb{R}_{++}$, that

$$\hat{\mathcal{I}}(\omega_j) \sim \frac{\sigma^2}{2}\chi_2^2 \sim \text{exponential}(2\pi s_y(\omega_j)), \quad \text{independent across } j > 0.$$

This turns out to hold *asympt'ly* for a large class of processes $\{y_t\}$.

Periodogram: asymptotic distribution (cont.)

Proposition (Brockwell & Davis, 1991, Thm. 10.3.2)

Consider a two-sided linear filter $y_t \equiv \Theta(L)u_t$ with abs. summ. $\Theta(L)$ and $u_t \overset{iid}{\sim} WN(0, \sigma^2)$. Assume $s_y(\omega) > 0$ for all $\omega \in [0, \pi]$.

For any m fixed freq's

$$0 < \omega^{(1)} < \cdots < \omega^{(m)} < \pi,$$

the vector

$$\left(\hat{\mathcal{I}}(g_T(\omega^{(1)})), \ldots, \hat{\mathcal{I}}(g_T(\omega^{(m)}))\right)'$$

converges in distribution to a vector of independent exponentially distributed rv's, where the j-th component has mean $2\pi s_y(\omega^{(j)})$.

- Intuitively, the periodogram interacts *approximately* multiplicatively with linear filters, like the spectrum.

Periodogram: smoothing

- We saw that each periodogram ordinate is inconsistent.
- But the previous theorem told us that

$$\tfrac{1}{2\pi}\hat{\mathcal{I}}(\omega_j) \approx s_y(\omega_j)(1 + U_j),$$

where U_j has mean zero, var. 1, and is approx. uncorrelated across j.

- By the LLN, we expect $\frac{1}{1+2B}\sum_{|k|\leq B} s_y(\omega_{j+k})U_{j+k} \approx 0$ for large B.
- Suggests a moving average estimator of $s_y(\omega_j)$:

$$\frac{1}{2\pi(1+2B)}\sum_{|k|\leq B}\hat{\mathcal{I}}(\omega_{j+k}) \approx \frac{1}{(1+2B)}\sum_{|k|\leq B} s_y(\omega_{j+k}) \approx s_y(\omega_j),$$

where the last approximation is valid if B/T is not too large and $s_y(\cdot)$ is cts (so $s_y(\omega_{j+B}) \approx s_y(\omega_j)$).

- Similar logic as in nonparametric (probability) density estimation!

Kernel spectral estimator

- More generally than the moving average, we will consider **kernel-weighted** spectral estimators

$$\hat{s}_y(\omega_j) \equiv \frac{1}{2\pi} \sum_{|k| \leq B} w_B(k) \hat{\mathcal{I}}(\omega_{j+k}),$$

$$w_B(k) \equiv \frac{w(k/B)}{\sum_{|m| \leq B} w(m/B)}, \quad |k| \leq B.$$

- The fct $w \colon [-1, 1] \to \mathbb{R}_+$ is called the **kernel**. We require it to be symmetric around 0, so that $\hat{s}_y(\cdot)$ is also symmetric. Note that $\hat{s}_y(\cdot)$ must also be nonnegative, since $w(\cdot) \geq 0$.
- The integer B is called the **bandwidth**. It determines how many ordinates we smooth across.
- Trade-off: large $B \Rightarrow$ high bias but low variance of $\hat{s}_y(\omega_j)$.
- Note: If $\omega_{j+k} \notin \mathcal{F}_T$, then replace with $\omega_{j+k} \pm 2\pi$.

Kernel spectral estimator: kernel and bandwidth choice

- Popular kernels:
  - Rectangular: $w(x) = 1$.
  - Triangular: $w(x) = 1 - |x|$.
  - Quadratic spectral (QS) or Epanechnikov: $w(x) = 1 - x^2$.
- In practice, the choice of kernel often matters less than the choice of bandwidth.
- We will talk about rules for bandwidth choice later in this course. Makes sense to check sensitivity to B. Some applications depend more on B than others ("does the spectrum have a spike?" vs. "is the spectrum broadly speaking decreasing in $\omega \in [0, \pi]$?").
- Think about what B and T imply for averaging across cycles of nearby frequencies $\omega_j = \frac{2\pi j}{T}$.

Kernel spectral estimator: asymptotic distribution

- Brockwell & Davis (1991), ch. 10.4–10.5, argue that, for $\omega \neq 0$,

$$\frac{\hat{s}_y(g_T(\omega)) - s_y(\omega)}{\sqrt{\sum_{|k| \leq B_T} w_{B_T}(k)^2}} \xrightarrow{d} N(0, s_y(\omega)^2) \quad \text{as } T \to \infty,$$

  provided that $B \equiv B_T \to \infty$ and $B_T/T \to 0$ as $T \to \infty$, $\int_{-1}^{1} w(x)^2 \, dx < \infty$, and regularity conditions hold.

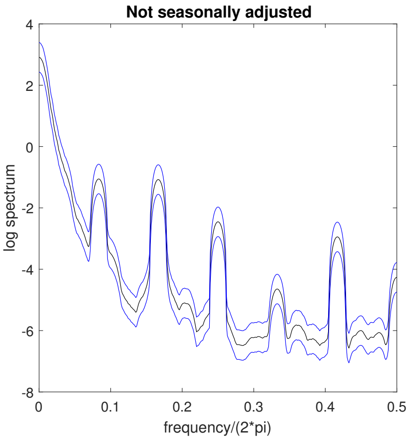
- The delta method implies a limit for the log spectrum:

$$\frac{\log \hat{s}_y(g_T(\omega)) - \log s_y(\omega)}{\sqrt{\sum_{|k| \leq B_T} w_{B_T}(k)^2}} \xrightarrow{d} N(0, 1) \quad \text{as } T \to \infty.$$

- Can construct pointwise conf. band using normal critical values.

- The above results require $\{y_t\}$ to have mean zero. If that is not the case, everything goes through if we simply replace $\hat{\mathcal{I}}(0)$ by $\hat{\mathcal{I}}(\omega_1)$ whenever the former appears in formulas.

Example: Kernel conf. band for unempl. rate spectrum



Sample: 1948:1–2017:8 (U.S.). Pointwise confidence level: 95%.
Kernel: Epanechnikov. Bandwidth: 10 ordinates.

Multivariate spectral representation theorem

Proposition (Brockwell & Davis, 1991, Cor. 11.8.1)

If $\{y_t\}$ is n-dim. cov. stat. with mean zero, there exists a complex-valued n-dim. orthogonal increment process $Z(\cdot)$ on $[-\pi, \pi]$ such that $Z(-\pi) = 0$ and

$$y_t = \int_{-\pi}^{\pi} e^{i\omega t}\, dZ(\omega) \quad \text{for all } t.$$

- An n-dim. complex orthog. incr. process $Z(\cdot)$ satisfies:
  - i) $E(Z(\omega)) = 0_{n\times 1}$, $E(Z(\omega)Z(\omega)^*)$ finite for all ω.
  - ii) $E[(Z(\omega_4) - Z(\omega_3))(Z(\omega_2) - Z(\omega_1))^*] = 0_{n\times n}$ if $[\omega_1, \omega_2] \cap [\omega_3, \omega_4] = \emptyset$.
  - iii) $E[(Z(\omega + \delta) - Z(\omega))(Z(\omega + \delta) - Z(\omega))^*] \to 0_{n\times n}$ when $\delta \to 0$.
- As in the univariate case, we have the intuitive interpretation

$$\int_{-\pi}^{\pi} e^{i\omega t}\, dZ(\omega) \approx \sum_{j=0}^{N-1} e^{i\omega_j t}[Z(\omega_{j+1}) - Z(\omega_j)], \;\; \omega_j \equiv \frac{2\pi j}{N} - \pi.$$

Spectral density matrix function

Definition

Let $\{y_t\}$ be an n-dim. cov. stat. time series with abs. summable ACF. The **spectral density matrix function** is given by

$$s_y(\omega) = \frac{1}{2\pi} \sum_{\ell=-\infty}^{\infty} e^{-i\omega\ell} \Gamma_y(\ell), \quad \omega \in [-\pi, \pi].$$

- $s_y(\cdot)$ is $n \times n$. The diagonal consists of the (real) univariate spectral densities $s_{y_j}(\cdot)$, $j = 1, \ldots, n$. Off-diagonal generally complex.
- Properties:
  - $s_y(\omega) = s_y(\omega)^*$ for all ω (asterisk: complex conjugate transpose).
  - $s_y(-\omega) = \overline{s_y(\omega)}$ for all ω.
  - $x^* s_y(\omega) x \geq 0$ for any $x \in \mathbb{C}^n$ and any ω.
- Inversion formula: $\Gamma_y(\ell) = \int_{-\pi}^{\pi} e^{i\omega\ell} s_y(\omega)\, d\omega$.

Spectral density matrix function: cross-spectrum

- Decompose $y_t = (q_t', r_t')'$, where $n_q \equiv \dim(q_t)$ and $n_r \equiv \dim(r_t)$ satisfy $n_q + n_r = n$. We write

$$s_y(\omega) = \begin{pmatrix} s_q(\omega) & s_{qr}(\omega) \\ s_{rq}(\omega) & s_r(\omega) \end{pmatrix}.$$

- The upper right block (dimension $n_q \times n_r$) is called the **cross-spectrum** between $\{q_t\}$ and $\{r_t\}$:

$$s_{qr}(\omega) = \frac{1}{2\pi} \sum_{\ell=-\infty}^{\infty} e^{-i\omega\ell} \operatorname{Cov}(q_t, r_{t-\ell}).$$

- Generally a complex matrix.
- Note: $s_{rq}(\omega) = s_{qr}(\omega)^*$.

Spectral density matrix function: cross-spectrum (cont.)

- Consider the bivariate case $y_t = (q_t, r_t)'$, $n_q = n_r = 1$.
- The **coherency** btw. q_t and r_t is the frequency-by-frequency analogue of the correlation coefficient:

$$K_{qr}(\omega) \equiv \frac{s_{qr}(\omega)}{\sqrt{s_q(\omega) s_r(\omega)}}, \quad \omega \in [-\pi, \pi].$$

- Define the BLP of q_t given all leads/lags of r_t:

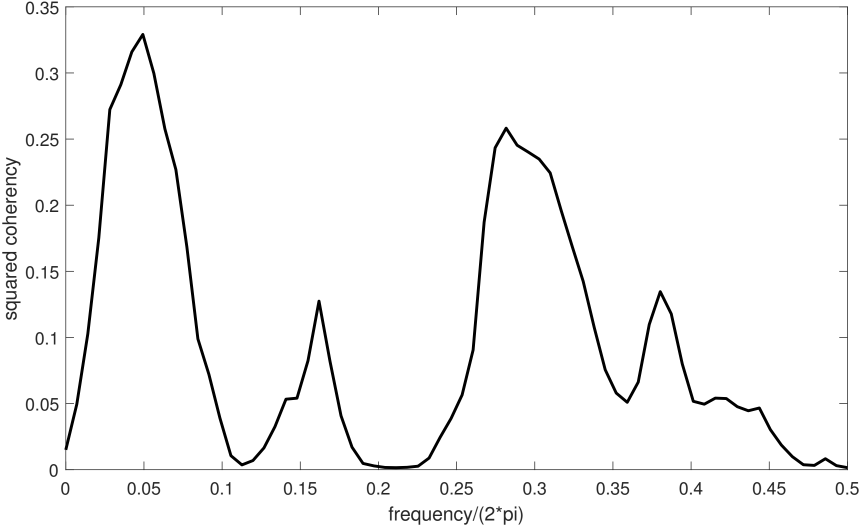
$$q_t^\dagger \equiv E^*(q_t \mid \{r_\tau\}_{-\infty<\tau<\infty}).$$

- One can show (see Problem Set 2) that

$$|K_{qr}(\omega)|^2 = \frac{s_{q^\dagger}(\omega)}{s_q(\omega)}, \quad \omega \in [-\pi, \pi].$$

  So the *squared coherency* has an R^2 interpretation: It's the fraction of the variance of frequency-ω cycles of $\{q_t\}$ that is explained by the best-fitting two-sided linear filter in $\{r_t\}$.

Example: sq. coherency btw. RGDP growth & term spread



Sample: 1982q1–2017q2 (U.S.). Kernel: Epanechnikov. Bandwidth: 8 ordinates.

Spectral density matrix function: linear filters

Proposition (Brockwell & Davis, 1991, Thm. 11.8.3)

Let $\{x_t\}$ be an n_x-dim. cov. stat. series with spectral density matrix fct $s_x(\cdot)$. Let $\Theta(z) = \sum_{\ell=-\infty}^{\infty} \Theta_\ell z^\ell$ be an abs. summ. $n_y \times n_x$ matrix polynomial. Then the spectral density matrix fct of the n_y-dim. series $y_t \equiv \Theta(L)x_t$ is given by

$$s_y(\omega) = \Theta(e^{-i\omega})s_x(\omega)\Theta(e^{-i\omega})^*, \quad \omega \in [-\pi, \pi].$$

- Proof is the same as in the univariate case.
- Exercise: What is the spectral density matrix fct of an n-dim. causal VARMA(p, q) process?
- Exercise: Let $y_t = \Theta(L)u_t$ be the Wold decomp. of a purely nondeterministic series. What is the cross-spectrum $s_{yu}(\omega)$?

Spectral density matrix function: estimation

- For estimation, could approximate the spectral density matrix fct using a VAR(p). S.e. by delta method or Bayesian approach.
- Alternatively, multivariate analogue of kernel spectral estimation:

$$\hat{s}_y(\omega_j) = \frac{1}{2\pi} \sum_{|k| \le B_T} w_{B_T}(k) \hat{\mathcal{I}}(\omega_{j+k}),$$

where we define the multivariate periodogram

$$\hat{\mathcal{I}}(\omega_j) \equiv \frac{1}{T} \left(\sum_{t=1}^{T} e^{-i\omega_j t} y_t \right) \left(\sum_{t=1}^{T} e^{-i\omega_j t} y_t \right)^* .$$

Kernel could be a scalar or could be a diagonal matrix fct. Details and limit theory in Brockwell & Davis (1991), ch. 11.7.

Whittle likelihood function

- Log likelihood of n-dim. mean-zero stat. *Gaussian* time series $\{y_t\}$ with spectral density matrix fct $\omega \mapsto s_y(\omega;\theta)$, which depends on parameters θ (e.g., spectrum implied by DSGE model):

$$L_T(\theta) = -\tfrac{nT}{2}\log(2\pi) - \tfrac{1}{2}\log\det(\Gamma_T(\theta)) - \tfrac{1}{2}y_T'\Gamma_T(\theta)^{-1}y_T,$$

  where $y_T \equiv (y_1', \dots, y_T')'$, and $\Gamma_T(\theta)$ is the $nT \times nT$ var-cov matrix of y_T, which is a function of $s_y(\cdot;\theta)$.

- $L_T(\theta)$ can be approximated by the **Whittle log likelihood function**

$$L_T^W(\theta) \equiv -nT\log(2\pi) - \tfrac{1}{2}\sum_{\omega_j \in \mathcal{F}_T}\left\{\log\det(s_y(\omega_j;\theta)) + \tfrac{1}{2\pi}\operatorname{tr}\left(\hat{\mathcal{I}}(\omega_j)s_y(\omega_j;\theta)^{-1}\right)\right\},$$

  where "tr$(\cdot)$" denotes the trace of a matrix. See Brockwell & Davis (1991), Prop. 4.5.2, for details on the validity of the approximation.

- In the univariate case $n = 1$, $L_T^W(\theta)$ corresponds to the asy. limit $\hat{\mathcal{I}}(\omega_j) \overset{iid}{\sim} \text{exponential}(2\pi s_y(\omega_j;\theta))$, $j \geq 0$.

Whittle likelihood function (cont.)

$$L_T^W(\theta) \equiv -nT\log(2\pi) - \frac{1}{2}\sum_{\omega_j \in \mathcal{F}_T} \left\{ \log\det(s_y(\omega_j;\theta)) + \frac{1}{2\pi}\operatorname{tr}\left(\hat{\mathcal{I}}(\omega_j)s_y(\omega_j;\theta)^{-1}\right)\right\}$$

- The Whittle (log) likelihood fct decomposes the Gaussian log likelihood into contributions across Fourier frequencies ω_j.
- In models where the spectrum $s_y(\omega;\theta)$ is quickly computable for any given θ, the Whittle likelihood is fast to compute (compute the periodogram using the discrete Fourier transform).
- If $\mu_y \neq 0$ possibly, just drop frequency 0 from the sum.
- The Whittle likelihood also suggests a strategy for determining how different frequencies of the data contribute to our results: Simply remove certain Fourier frequencies from the sum (e.g., low/high frequencies). Could even weight frequencies differentially.

Application: DSGE models in the frequency domain

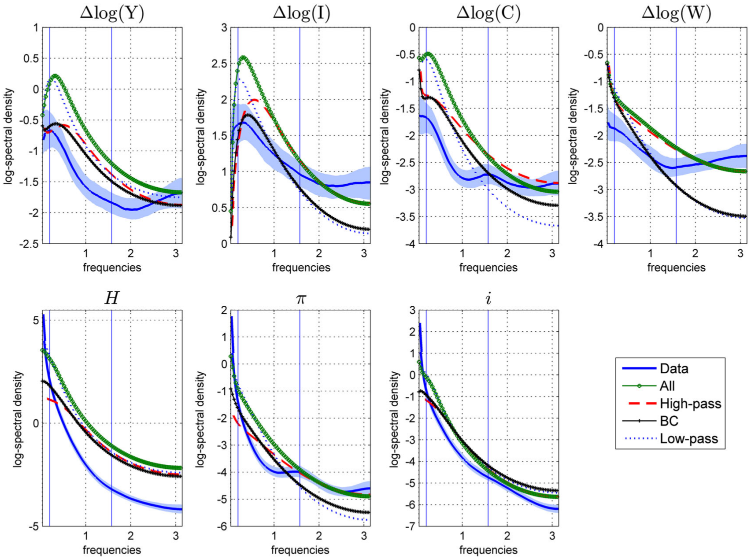


Figure 2. Log-spectra of data and models estimated on different frequency bands. Vertical bars separate the frequency domain in the three regions *low*, *BC* and *high*. *low* denotes cycles with period $32 < \text{per} < \infty$ quarters; *BC* denotes cycles with period $4 < \text{per} < 32$ quarters; *high* denotes cycles with period $2 < \text{per} < 4$. For details, see Appendix C (supporting information) and footnote 6

Source: Sala (2015)